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4 SEM MCA (NCBCS) 2

2015

(June)

DATABASE MANAGEMENT SYSTEM–II

Paper : 402

Full Marks : 75

Pass Marks : 34

Time : Three hours

The figures in the margin indicate full marks for the questions.

*Answer **any five** questions.*

1. *(a)* What do you understand by query processing and query optimization? Discuss. 5

   (b) Describe sort merge join algorithm. 5

   (c) What do you mean by cost estimation in query optimisation? Explain different cost component for query optimization. 2+3=5

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2. (a) What do you understand by query tree and query graph? Discuss. 3

(b) What is speed up? How scale up is different from it? Explain. 1+3=4

(c) Consider the following relations : 8

STUDENT (SNo, SNAME, DoB, ADDRESS, DNo, SEMESTER, GENDER)

PROJECT (PNo, PNAME, PTYPE, PLOCATION)

WORKS_ON (SNo, PNo, DATE)

Showing different stage of query optimization, find the name of the student born after 2000, who work on a project name "E_Learning".

3. (a) What is Inter operation parallelism? What is pipeline parallelism? Discuss its advantages. 1+2+2=5

(b) What is I/O parallelism? How can it be achieved using data partitioning technique? 1+4=5

(c) Explain database security in Oracle. 5

4. (a) Define data mining. What are the different application areas of data mining? 5

(b) Discuss the two phase commit protocol 5

(c) What do you mean by data warehousing? Explain. 5

5. (a) Discuss how objects are made persistent in object oriented database 5

(b) State the characteristics of an object identifier (OID) in an object oriented database. 5

(c) What is data replication? Write the advantage and disadvantage of data replication. 1+4=5

6. (a) What is locking? What are the two mechanisms for locking data in a database. 1+4=5

(b) What is spatial database? Write the feature of spatial database. 2+3=5

(c) What do you mean by main memory database system (MMDB)? What are the differences in the properties of main memory and disk resident database system? 1+4=5

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7. Write short notes on : ***(any three)*** 5×3=15

(a) Coordinator Selection

(b) Partitioned parallel hash join

(c) Graphical database

(d) Time stamping

(e) ROLAP vs MOLAP.

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4 SEM MCA (NCBCS) DBMS 2

2019

(June)

COMPUTER APPLICATION

Paper : MCA–402

(Database Management System–II)

Full Marks : 75

Pass Marks : 34

Time : Three hours

The figures in the margin indicate full marks for the questions.

*Answer **any five** questions.*

1. *(a)* EMP(SSN, ENAME, BDATE, ADD, SALARY, DEPT_NO, SEX)
DEPT (DNO, DNAME, MGR_NO., MGR_START_DATE)
PROJECT(PNUMBER, PNAME, PLOCATION, DNO)
WORKS ON(ESSN, PNO, HOURS)
From the above database, consider the query :
SELECT ENAME
FROM EMPLOYEE, WORKS_ON, PROJECT
WHERE PNAME='ROBOTICS' AND
PNUMBER=PNO AND ESSN=SSN AND
BDATE>'1980-10-13';
Draw the initial query tree and show how the query tree is optimized. 7

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(b) A file of 1024 blocks is to be sorted with an available buffer space of 5 blocks. How many passes is needed in the merged phase of external sort merge algorithm ? 4

(c) Explain Query processing and query optimization in details. 4

2. (a) Explain different steps to process a high level query. 5

(b) Explain *at least one* algorithm used for processing of join operation. 5

(c) Describe Semantic query optimization. How does it differ from other query optimization technique ? 5

3. (a) Draw the block diagram of Shared disk multiple cpu and shared nothing multiple cpu parallel database architecture and give advantage and disadvantage of each. 3+3=6

(b) Define Database security. Explain the type of database security issues. 2+3=5

(c) Explain Authentication and Authorization in details. 2+2=4

4. (a) Explain Data fragmentation and Da replication with advantage an disadvantage. 4+4

(b) Explain in details Timestamp meth for distributed database.

(c) Explain the advantages of distribut database.

5. What are the usages of commit protocol Discuss how the following are handled i 2PC : 2+3+3

1. Failure of a participating site.

2. Failure of the coordinator.

(b) Define I/O parallelism. Explain Rour Robin partitioning technique wi advantage and disadvantage.

(c) Explain synchronization.

6. (a) Explain Intra Operation Parallelisn How is it different from Inter Operatio Parallelism ? Explain the forms of Int Operation Parallelism. 2+2+4

(b) Explain how a persistent point persists its identity.

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(c) Explain Object Query language (OQL). 2

7. Write short notes on : ***(any three)*** 3×5=15

(a) Data warehousing

(b) Main memory database

(c) Coordinator selection in distributed database

(d) Inter query parallelism

(e) Selectivity.

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4 SEM MCA (NCBCS) DBMS 2

2018

(June)

COMPUTER APPLICATION

Paper : 402

(Database Management System–II)

Full Marks : 75

Pass Marks : 34

Time : Three hours

The figures in the margin indicate full marks for the questions.

*Answer **any five** questions.*

1. *(a)* Discuss query processing and query optimization in details. 6

   (b) Explain Query tree and Query graph with examples. 6

   (c) Explain Sort-merge join algorithm. 3

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2. (a) Define Speed up and Scale up. Discuss the factors affecting Speedup and Scaleup. 3+3=6

(b) Explain the following search methods for simple selection. Give an expression for the estimation of cost in each case : 9

(1) Binary search

(2) Using a primary index to retrieve multiple records

(3) Using clustering index.

3. (a) Discuss the fragment and replicate join technique. 6

(b) What is IO parallelism ? How can it be achieved using data partitioning technique ? 7

(c) Describe the different stages of Knowledge discovery in database (KDD). 2

4. (a) Explain the 3-phase commit protocols with details. 5

(b) Discuss how objects are made persistent in object oriented database. 5

(c) Explain Database security in oracle. 5

5. (a) What do you mean by Inter Operat parallelism ? What is Independ parallelism ? Discuss its advantage

(b) Define Data Mining. What are different applications area of D mining ?

(c) Explain Metadata with its types.

6. (a) Explain the different models used parallel database architecture w advantages and disadvantages.

(b) Explain the Parallel External Sort Me algorithm.

7. Write short notes on : ***(any three)*** 5x3=

(a) Data Warehousing

(b) Concurrency control

(c) Deadlock handling in distribut database,

(d) Graphical database.

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4 SEM MCA DBMS 2 (NCBCS)

2016

(June)

COMPUTER APPLICATION

Paper : MCA-402

(Database Management System–II)

Full Marks – 75

Pass Marks – 34

Time – Three hours

The figures in the margin indicate full marks for the questions.

Answer any *five* questions.

1. (a) What do you understand by query tree and query graph ? Discuss. 3

   (b) Discuss at least four search algorithm for selecting records from a file. 2×4=8

   (c) Explain different steps to process a high level query. 4

2. (a) Explain at least one algorithm used for processing of Join operation. 5

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(b) Define speed up and scale up. Discuss the factors affecting speed up and scale up. 5

(c) Explain the effect of available buffer space and join selection factor on join operation with an example. 5

3. (a) Explain the time stamping method. 5

(b) Explain how data replication is useful in distributed DBMS. 5

(c) What is graphical database ? Write the feature of graphical database. 5

4. (a) What are the usages of commit protocol ? Discuss the three phase commit protocol. 7

(b) Discuss how failure handling is performed in two phase commit protocol. 8

5. (a) What is concurrency control ? Explain the single-lock manager approach with its advantage and disadvantage. 2+6=8

(b) What do you mean by intraquery parallelism ? Describe two algorithms to implement intraquery parallelism. 1+6=7

6. (a) What is knowledge discovery datab (KDD) ? Discuss the different stages KDD.

(b) What is object identity ? How objects represented in object oriented datal management system ? Discuss.

7. Write short notes on any *three* : 3×5

(a) Independent parallelism

(b) Coordinator selection

(c) I/O parallelism

(d) Parallel nested loop join

(e) Metadata.

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Total number of printed pages–4

4 SEM MCA DBMS 2

2017

(June)

COMPUTER APPLICATION

Paper : 402

(Database Management System–II)

Full Marks : 75

Pass Marks : 34

Time : Three hours

The figures in the margin indicate full marks for the questions.

*Answer **any five** questions.*

1. *(a)* Discuss the reason for converting SQL queries into relation algebra queries before optimization is done. 5

   (b) What is meant by the term Heuristic optimization ? Discuss the main heuristics that applied during query optimization. 5

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(c) What is meant by Semantic query optimization ? How does it differ from other query optimization technique ? 5

2. (a) What do you meant by query tree and query graph ? How does a query tree represent a relation algebra expression ? Discuss the rules for transformation of query tree and identify when each rule should be applied during optimization. 10

(b) Explain different cost component for query execution. 5

3. (a) Describe parallel external hash join. 5

(b) Explain shared memory architecture with advantage and disadvantage. 5

(c) What do you meant by I/O parallelism ? Explain Range partitioning technique with advantage and disadvantage. 5

4. (a) What is *two* Phase commit protocol ? List the control messages used in two Phase commit protocol. 5

(b) What is Deadlock ? List and discuss the *four* conditions for deadlock. 5

(c) What is a fragment of a relation ? What are the main types of fragment ? 5

5. (a) How does the concept of object in object oriented model differ from concept of entity in ER model ? 5

(b) Explain main memory database system (MMDB). 5

(c) What is data warehousing ? Explain. 5

6. (a) Distinguish between pipeline parallelism and independent parallelism. 6

(b) Explain database security in oracle. 4

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(c) Define Information retrieval system. Explain the objective of information retrieval system. 5

7. Write short notes on : ***(any three)*** 3×5=15

(a) Distribution transparency

(b) Sort merge join algorithm

(c) Selectivity

(d) Concurrency Control

(e) Inter query parallelism.

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